Community Perceptions of the China–Pakistan Economic Corridor: Implications for Economic, Social, and Sustainable Development

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Abstract

The China–Pakistan Economic Corridor (CPEC) is a flagship infrastructure initiative designed to enhance regional connectivity, economic development, and sustainable growth. This study examines how community perceptions of CPEC relate to perceived economic development (PED) and perceived societal development (PSD), and how these perceptions are associated with sustainable development and support for CPEC. Drawing on Social Exchange Theory, the study adopts a mixed-methods approach, combining survey data from 100 respondents with qualitative interviews from 15 participants representing urban, rural, and peri-urban communities in Pakistan. Quantitative data were analyzed through correlation, multiple regression, and bootstrapped mediation analyses, while qualitative data were examined using thematic analysis. The findings indicate that perceived economic development ($\beta = 0.612$, $p < .001$) and perceived societal development ($\beta = 0.222$, $p = .030$) are significant predictors of support for CPEC. Furthermore, sustainable development partially mediates these relationships, highlighting its role in translating infrastructure benefits into broader social and economic gains. Qualitative findings indicate perceived improvements in employment opportunities, transportation infrastructure, education, and healthcare services. However, participants also expressed concerns regarding unequal distribution of benefits, limited community inclusion, and potential environmental challenges. The study concludes that strengthening CPEC's long-term contribution to sustainable development requires inclusive policymaking, effective environmental management, and meaningful community participation. Because the design is cross-sectional and perceptual, the findings should be interpreted as associations rather than causal effects. The findings offer insights for policymakers seeking to align large-scale infrastructure projects with sustainable development objectives and the United Nations Sustainable Development Goals.

Keywords: Sustainable development, CPEC, Perceived Economic Development, Perceived Societal Development

1. Introduction

Development infrastructure, especially road and transportation, has long been a cornerstone of economic and social development. Transportation systems in developed nations are well developed, play a major role in economic prosperity, high living standards, and lowered environmental effects. These infrastructure systems help the smooth flow of goods and people, the development of industries, the availability of education and healthcare, and increased connectivity of the region and the world (Yuanchun et al., 2024). This is, however, not the case in many developing countries, especially in Pakistan. Poor infrastructure and ineffective transport systems are significant hindrances to development, restricting economic growth, hampering access to vital services, and aggravating environmental issues, especially in rural and urban society (Asif et al., 2024).

The CPEC, announced in 2013 as one of the flagship projects of the Belt and Road Initiative directed by the Communist Party of China (CPC), aims to address many of the development problems of Pakistan. CPEC is

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a massive infrastructure project, including new roads, railways, pipelines, and energy projects linking Chinese western provinces to Gwadar port in Pakistan (Hussain et al., 2024a). This initiative is expected to benefit Pakistan through improved connectivity, access to more international markets, and substantial improvements in domestic infrastructure, such as roads, bridges, and power plants (Zhang, 2023). The scale and scope of CPEC offer considerable potential to boost the economy and create jobs and improve overall living standards for ordinary people (Kanwal et al., 2022). However, the success of CPEC is measured not only in terms of infrastructural and economic development but also in terms of sustainable development, which is crucial for long-term environmental and social sustainability.

Sustainable development, as defined in international policies such as the United Nations Sustainable Development Goals (SDGs), refers to development that meets present human needs while considering the Earth's carrying capacity to support future generations (United Nations, 2022). CPEC is aligned with SDG 9 (Industry, Innovation, and Infrastructure) insofar as it provides sustainable, efficient, and resilient infrastructure (Asif et al., 2024). The impact of CPEC on specific SDGs, namely SDG 8 Decent Work and Economic Growth, and SDG 13 Climate Action, is also relevant because these goals raise issues of equity in economic development, the social cost of economic activities, and climate change mitigation (Bashir et al., 2022). To assess the degree to which CPEC is perceived to contribute to sustainable development, it is useful to focus on the level of Perceived Economic Development (PED) and Perceived Societal Development (PSD) among the local population. Such perceptions are important indicators of how projects like CPEC are evaluated regarding their sustainability.

Perceived Economic Development (PED) is the extent to which people believe that their nation has made economic progress in terms of employment opportunities, income growth, and quality infrastructure (Ahmad et al., 2025). Within the frame of CPEC, PED reflects the perception of the Pakistani population toward the economic transformations associated with the project, such as employment and trade connectivity. Assessing PED therefore provides an indication of the real-world applicability of infrastructure projects, since economic challenges in developing nations have been deeply linked to both social and ecological variables (Shahzad et al., 2023).

Perceived Societal Development (PSD) extends the idea of development to societal betterment. PSD measures how the local population perceives changes in quality of life, including living standards, healthcare, education, employment, and overall well-being (Kanwal et al., 2020). In the case of CPEC, PSD captures the perception of Pakistani people regarding the effects of the project on their individual and societal welfare, both economically and socially. PSD is conceptually linked to SDG 3 (Good Health and Well-Being), SDG 4 (Quality Education), and SDG 10 (Reduced Inequalities) because the presence of CPEC infrastructure may increase access to necessary services, improve quality of life, and decrease inequalities (Anwar et al., 2022). As a result, the perceptions of host communities toward CPEC are crucial in assessing the socio-economic effects of the project and its alignment with the SDGs.

CPEC, PED, and PSD are connected with sustainable development, which is important for the future growth of infrastructure projects. Sustainable development in Pakistan requires that economic growth, societal uplift, and environmental conservation under the CPEC umbrella be integrated. In particular, the evaluation of CPEC should consider the 12th SDG on Responsible Consumption and Production and the 13th SDG on Climate Action because the economic benefits obtained in the framework of cooperation may also affect the environment (Balsalobre-Lorente et al., 2023). Through sustainable development, Pakistan may transition into a low-carbon economy by creating green infrastructure, renewable energy projects, and a low-carbon transport system under CPEC (Bashir et al., 2022).

Existing research on CPEC has largely examined economic and infrastructural outcomes, and several studies have used Social Exchange Theory to explain community support. However, three gaps remain. First, few studies simultaneously model the economic and societal dimensions of perceived development as parallel predictors within a single framework. Second, the mechanism through which these perceptions are associated with support for CPEC namely, sustainable development has rarely been tested empirically. Third, most studies rely on either quantitative or qualitative evidence, whereas the present study integrates both to contextualize the statistical findings. This study therefore makes three contributions: (a) it specifies and tests a mediation model in which sustainable development partially mediates the relationship between PED/PSD and support for CPEC; (b) it combines survey data with interview evidence to explain why the statistical relationships take the form they do; and (c) it applies Social Exchange Theory to derive the hypotheses rather than using it only as background.

Based on these relationships, the study provides information on how massive infrastructure projects like CPEC can be successfully managed to produce sustainable results. To fulfill this purpose, the study addresses the following research questions:

RQ1: How are community perceptions of CPEC associated with Perceived Economic Development (PED) and Perceived Societal Development (PSD) among local communities?

RQ2: How are PED and PSD associated with Sustainable Development and, in turn, with Support for CPEC?

The research promotes balanced economic and social growth without necessarily affecting environmental integrity, thus supporting the SDGs. The information below is meant to guide policymakers in making development policies more responsive to the perceptions of local communities. Furthermore, the study provides recommendations on the efficiency of infrastructure spending for creating economic value, enhancing social welfare, and reducing negative environmental impacts. Finally, this research emphasizes the importance of perception-based assessments in designing policies for achieving sustainable development and equity for sustainable success. The remainder of the paper is organized as follows: Section 2 explains the literature review; Section 3 describes the theoretical framework and hypotheses. Section 4 discusses the data and methodology; Section 5 explains the results; Section 6 discusses the findings; and Section 7 explains the conclusion and policy recommendations.

2. Literature Review

2.1 Social Exchange Theory and Its Application in Infrastructure Development

Social Exchange Theory (SET) is among the most widely used paradigms for analyzing community attitudes toward development projects such as CPEC. SET holds that community members evaluate their judgments according to the perceived utility and sacrifice associated with an exchange. In the light of CPEC, SET implies that local communities will support projects if they perceive economic gains in terms of economic development, employment opportunities, and infrastructure development, while simultaneously weighing costs that may involve social or environmental implications (Arshad et al., 2024). Although a substantial body of research has explored community support for CPEC using SET, the focus has generally been on utilitarian or tangible gains, such as economic development, without much regard to social or cultural gains such as enhanced social capital and cultural transformation (Hussain et al., 2024b). (Abbas et al., 2025) found that perceived benefits of infrastructure projects are positively related to community support. Higher-income groups may obtain concrete economic gains from infrastructure projects, while

lower-income or marginalized groups may perceive little improvement, resulting in different observations of the project's success. The current literature also lacks a sufficiently strong distinction between perceived benefits and risks.

This difference between perceived benefits and actual impacts opens a gap in understanding how Perceived Economic Development (PED) differs from actual economic development. Whereas PED is based on people's perception of economic growth, actual economic development concerns measurable outcomes such as GDP growth rate or employment status (Crisp et al., 2024). This distinction is important because people may approve of a project based on beliefs about growth even when actual results are more modest. This gap must be addressed to assess the effectiveness of mega-construction projects.

2.2 Perceived Economic Development (PED) in the Context of CPEC

Perceived economic development (PED) may be described as people's perception of the economic impact of a project on employment, income, and quality of facilities. In the context of CPEC, education, media, and direct experience of infrastructure projects can strongly influence PED. Azhar et al., (2025) noted that PED factors are measured at an objective level by economic activities (employment, income growth) and at the perceived level by members of the respective communities, and that these may not coincide. (Zou et al., 2022) showed that those perceiving closer proximity to infrastructure development projects were more likely to perceive economic benefits than those perceiving lower proximity. This confirms the geographic distribution of PED perceptions. CPEC brings about improved infrastructure and connectivity, but those who do not directly witness such changes may be more critical of the project. In addition, Becker and Woessmann, (2024) noted that education and media correlate with economic development assessment. Overall, the level of awareness regarding the possible benefits of CPEC appears to have a positive relationship with perceptions of the project. This dynamic emphasizes the role played by perceived information quality in the formation of PED, meaning that media and education may play an intermediary role in shaping public opinion on large-scale infrastructure projects. However, there are still research gaps concerning how perceptions of economic benefits are linked with the actual development impact of PED. At the same time, (Shahzad et al., 2023) state that CPEC can increase Pakistan's overall GDP, and (Zhang et al., 2021) observe that there may be long-term growth imbalance risks. The literature is also unclear about whether people in marginalized areas hold perceptions similar to those in big cities regarding the level of benefits they expect to accrue or their perception toward CPEC. Addressing these gaps is important for understanding the social and economic effects of CPEC and similar projects in other areas.

2.3 Perceived Societal Development (PSD) and Its Role in CPEC

Economic development is considered an important criterion in evaluating the effectiveness of infrastructure projects, but another criterion is perceived societal development (PSD). PSD captures how people and societies judge enhancements in their social conditions, such as healthcare, schools, and jobs (Ullah et al., 2024). Besides economic development, CPEC, through its pace of infrastructure development, also aims to improve social development and standards for better service provision.

PSD is a broader development concept than GDP or gross domestic product size. (Anwar et al., 2024) define PSD as the improvement of objective conditions (the availability of better healthcare centers and schools) and the psychological perception of the well-being and capability of the community. In the case of CPEC, people in these communities may perceive a change in their societal well-being, such as an increase in the standard of living, healthcare, and education, even though these may not be captured in statistical data

(Sulaiman, 2024). However, there is a significant research limitation: constructs such as Perceived Societal Development (PSD) have not been adequately specified. While several papers refer to PSD, few provide a clear, operational definition of what PSD means or develop a reliable and valid way of assessing it. Consequently, there is ambiguity in how PSD is defined and assessed in various settings. According to (Hussain et al., 2025), PSD is a function of both personal characteristics (income and education) and contextual characteristics (community togetherness and local power). However, the measurement and definition of these variables need further development in present-day research, which creates challenges in ascertaining and comparing the impacts of CPEC on societal advancement.

In addition, the distinction between tangible and non-tangible social values, such as improvements in infrastructure and the social and psychological well-being of individuals or the sense of community and belongingness, needs to be better defined in the literature. Sharper lines must be drawn to clarify how CPEC infrastructure affects communities (McLaughlin, 2024). Although communities might perceive improvements in living standards through better roads and utilities, the real empowerment or agency of these populations may be more subtle and requires a detailed qualitative investigation.

2.4 Gaps in Existing Research and Areas for Future Exploration

Table 1. Summary table of key findings from existing research

Study	Focus	Key findings	Gaps in literature
Kanwal et al. (2020)	Perceived benefits of CPEC on local communities	Positive perceptions of CPEC's benefits, including economic growth and job creation	Lack of focus on intangible benefits like empowerment
Ali et al. (2018)	Public perception of infrastructure development	Local communities perceive improvements in infrastructure quality and living standards	No differentiation between tangible and intangible benefits
Bashir et al. (2022)	CPEC and renewable energy development	CPEC has the potential to promote low-carbon energy solutions, supporting Pakistan's transition to a sustainable economy	Insufficient focus on the integration of sustainability practices
Anwar et al. (2022)	Societal development and public perception of CPEC	CPEC has the potential to improve living standards, healthcare, and education	Lack of clear operational definitions for Perceived Societal Development

Note: The Sohail & Chen (2022) entry has been removed because the source was retracted. See the note in the reference list.

3. Theoretical Framework and Hypotheses [R2-2]

This study draws on Social Exchange Theory (SET) to explain the relationships among Perceived Economic Development (PED), Perceived Societal Development (PSD), Sustainable Development (SD), and Support for CPEC. SET was introduced by Homans in 1958 and holds that people and communities evaluate the worth of interactions and relationships in terms of gains and losses. The theory is useful for explaining how community-level stakeholders view and engage with large infrastructure initiatives such as CPEC.

Community stakeholders are likely to appraise the project based on perceived benefits from the project (economic development, employment opportunities, social transformation) and perceived costs incurred by the local communities (for instance, loss of arable land and social unrest). The more benefits are perceived relative to costs, the more likely they are to support and participate in the project (Lehtinen and Aaltonen, 2024).

From theory to hypotheses. SET generates the study's hypotheses in three steps. First, if community members perceive that CPEC produces economic gains for them (PED), the perceived benefits of the exchange increase, which should raise support for the project. Second, if community members perceive that CPEC produces social gains (PSD) better health, education, and welfare the perceived benefits of the exchange also increase, which should further raise support. Third, if community members perceive that CPEC contributes to sustainable development (i.e., that benefits are not achieved at unacceptable environmental or intergenerational cost), the perceived net benefit of the exchange increases, and support should rise. The mediation model follows from the same logic: PED and PSD are perceptions of the benefits of the exchange, SD is the perception that those benefits are sustainable, and support for CPEC is the behavioral outcome of the exchange evaluation.

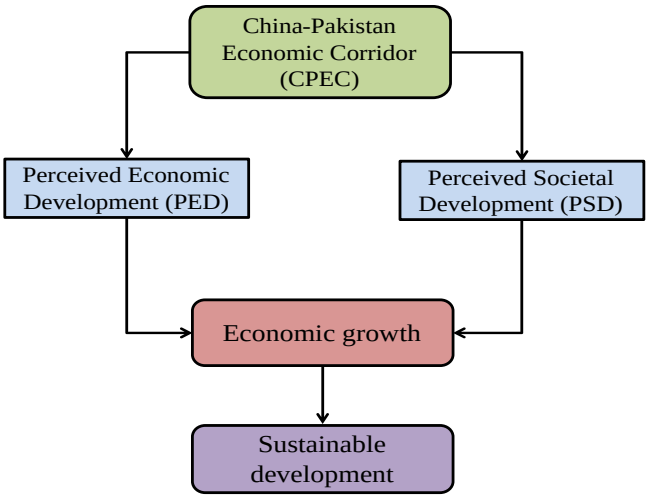


Figure 1: Theoretical model

In SET, backing for CPEC is rooted in the assessment of the utility that it offers to people in the region. PED concerns how people perceive CPEC affecting their livelihood. Better roads and transportation systems, more employment, and better job opportunities are related to SDG 8. The perceived success of these benefits is expected to increase backing for the project and help ensure continued economic profitability (Shahzad and Sunawar, 2024).

Unlike CPEC's economic returns, PSD concerns the social returns in enhanced health, education, and social welfare. According to SET, if the perceived social returns of CPEC exceed the possible social costs, such as displacement or cultural impacts, communities will be more likely to support the project. This perception of societal improvement relates directly to SDG 3 (Good Health and Well-Being), SDG 4 (Quality Education),

and SDG 10 (Reduced Inequalities). Education and healthcare are among the social infrastructures that enhance the quality of society and, therefore, support for the project (Waheed et al., 2024).

CPEC also incorporates green development features, including renewable energy and green buildings. From the perspective of SET, communities are willing to support infrastructural projects if they believe that the benefits accruing to the environment exceed the costs. If local communities perceive that CPEC contributes positively to sustainable development through reduced carbon emissions from renewable energy and better environmental protection (Bensadi, 2024), the project should receive more support. This corresponds to SDG 13 (Climate Action).

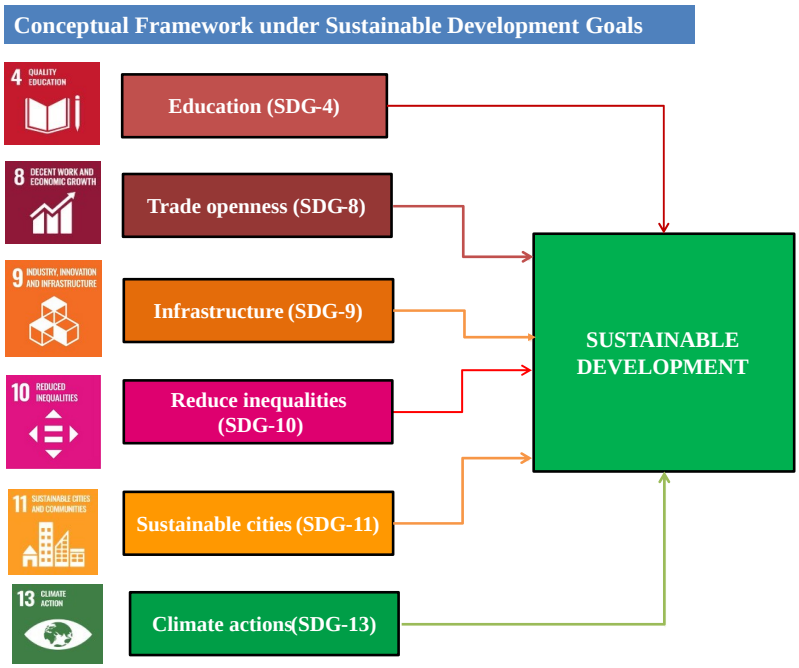


Figure 2: Conceptual framework

Support for CPEC is expected to increase when both Perceived Economic Development (PED) and Perceived Societal Development (PSD) are favorable. SET postulates that when the project enjoys social support, local communities assess the economic and social costs to be outweighed by benefits. This, in turn, supports the adoption of the project and its long-term sustainability (Asif et al., 2025). The integration of the economic and social dimensions of development also contributes to SDG 10 (Reduced Inequalities) and SDG 11 (Sustainable Cities and Communities), making CPEC a potential benchmark for inclusive and sustainable development.

The SET framework links PED and PSD to sustainable development and shows how community perceptions shape responses to large-scale infrastructure projects like CPEC (Aziz et al., 2025).

Hypotheses. Based on the preceding theoretical discussion, the following hypotheses are proposed and numbered consistently throughout the manuscript:

- H1: Perceived Economic Development (PED) is positively associated with Support for CPEC.
- H2: Perceived Societal Development (PSD) is positively associated with Support for CPEC.
- H3: Sustainable Development (SD) mediates the relationship between PED and Support for CPEC.

- H4: Sustainable Development (SD) mediates the relationship between PSD and Support for CPEC.
- H5: PED is positively associated with Sustainable Development (SD).
- H6: PSD is positively associated with Sustainable Development (SD).

4. Data and Methodology

4.1 Data Collection

This study evaluated local community perceptions of CPEC and their association with Perceived Economic Development (PED) and Perceived Societal Development (PSD). The results are based on quantitative survey data from 100 participants and qualitative interviews with 15 participants in various regions of Pakistan, as shown in Table 2. The findings are discussed in relation to the SDGs, specifically SDG 8 (Decent Work and Economic Growth), SDG 9 (Industry, Innovation, and Infrastructure), SDG 10 (Reduced Inequalities), SDG 13 (Climate Action), and SDG 3 (Good Health and Well-Being).

Sampling and geographical coverage. Participants were recruited using a combination of purposive and snowball sampling. The target population consisted of adults (aged 21 years and above) residing in communities directly affected by CPEC infrastructure projects. Eligibility criteria were: (a) residence in a CPEC-affected district for at least two years; (b) age 21 or above; and (c) willingness to provide informed consent. The sample was distributed across three settings: urban industrial zones (n = 40), rural agricultural areas (n = 35), and peri-urban communities (n = 25). This distribution was chosen to allow comparison of urban, rural, and peri-urban experiences, which prior research has identified as an important source of variation in CPEC perceptions (Zou et al., 2022). The sample size of 100 was determined by resource constraints and is consistent with the minimum requirements for detecting medium-to-large effects in multiple regression with two predictors (Cohen, 1992; Green, 1991). Because the sample is non-probabilistic, the findings are not statistically generalizable to the entire Pakistani population; they are analytically generalizable to the theoretical relationships under study.

4.2 Measurement Instruments

This study assessed Perceived Economic Development (PED) using eleven items adapted from (Kanwal et al., 2019). Perceived Societal Development (PSD) was measured using eleven items modified from (Lee et al., 2018). Support for CPEC was assessed using eight items that were used in previous studies (Ali et al., 2018; Kanwal et al., 2019). Sustainable Development was measured using eleven items adapted from (Kanwal et al., 2019) and Karim (2022). Every measure was provided using a five-point Likert scale ranging from strongly disagree to strongly agree. Demographic variables, including age, gender, education level, and experience, were also included.

4.3 Qualitative Component

To complement the quantitative survey findings, qualitative data were collected to gain deeper insights into community perceptions regarding CPEC's socio-economic and environmental impacts. Semi-structured open-ended questions were incorporated into the survey instrument, allowing respondents to elaborate on their experiences and concerns related to CPEC-related developments.

Participant selection. A purposive subsample of 15 respondents was selected for brief face-to-face interviews. The subsample comprised 5 participants from urban industrial zones, 5 from rural agricultural

areas, and 5 from peri-urban communities directly affected by CPEC infrastructure projects. Participants were selected to ensure representation across gender, age, and education levels.

Interview procedure. Interviews lasted approximately 20–30 minutes each. The interview guide covered four domains: (a) perceived economic impacts of CPEC; (b) perceived social impacts; (c) perceived environmental impacts; and (d) views on community participation and benefit distribution. Questions were open-ended and followed up with probes.

Recording and transcription. With participants' consent, interviews were audio-recorded and transcribed verbatim. Transcripts were checked against recordings for accuracy. Where consent to record was not given, detailed notes were taken during and immediately after the interview.

Coding and theme development. Transcripts were coded and analyzed using thematic analysis following the procedures recommended by Braun and Clarke (2006). The analysis followed six phases: familiarization, initial coding, searching for themes, reviewing themes, defining and naming themes, and reporting. Two researchers independently coded the transcripts; discrepancies were resolved through discussion. This approach enabled the identification of recurring themes, including job sustainability concerns, unequal benefit distribution, healthcare accessibility gaps, and environmental degradation. The qualitative findings are presented as a coherent analytical section (Section 5.5) and are integrated with the quantitative findings in the Discussion.

Ethics. Participation was voluntary, and all participants provided informed consent before taking part. Participants were informed of the purpose of the study, the voluntary nature of participation, and their right to withdraw at any time without consequence. Confidentiality and anonymity were maintained: no identifying information was collected, and participants are referred to by anonymized identifiers (e.g., Urban-1, Rural-2). Interview recordings and transcripts were stored securely and accessible only to the research team. The study procedures were reviewed and approved by the relevant institutional review committee prior to data collection.

Table 2. Demographic information of respondents

Demographics	**Frequency**	**Percentage**
Gender		
Male	50	50
Female	50	50
Age		
21–30 years	71	71
31–40 years	24	24
41–50 years	5	5
Education		
Graduation	24	24
Master	55	55
Ph.D.	19	19
Others	2	2
Experience		
< 5 years	75	75
6–10 years	23	23
11–15 years	2	2

Source: survey data

Note: The sample distribution across urban (n = 40), rural (n = 35), and peri-urban (n = 25) settings is reported in Section 4.1.

5. Statistical Analysis and Results

To directly address the research questions posed in this study, the following statistical analyses examine the association between CPEC perceptions and perceived economic and societal development (RQ1), and the subsequent association of these perceptions with sustainable development and support for CPEC (RQ2).

5.1 Descriptive Statistics and Alpha Reliability

Descriptive statistics and Cronbach alpha were performed to investigate the internal consistency of the measurement tools and to describe the distribution of variables. The analysis of descriptive statistics provided information about the distribution of variables. In general, descriptive statistics shed light on the distribution and nature of variables in a dataset (George and Mallery, 2024). Table 3 shows the alpha reliability and descriptive statistics of the study's constructs. The analysis indicated that reliability was high for Perceived Economic Development (.87), Perceived Societal Development (.86), Support for CPEC (.90), and Sustainable Development (.95). These alpha values suggest that the measurement instrument has high internal consistency. Table 3 also reports descriptive statistics for each of the constructs: minimum, maximum, number of items (K), mean (M), and standard deviation (SD). The average scores were 32.35 and 45.52, with standard deviations between 4.62 and 7.19.

5.2 Correlation Analysis

The correlation analysis was used to examine the relationships among variables of the study: PED, PSD, Support for CPEC, and sustainable development. To test the direction and strength of these relationships, Pearson correlation coefficients were obtained, making it possible to test the hypotheses empirically. The results of the correlation are given in Table 4, which shows strong positive relationships.

PED was positively and significantly correlated with PSD ($r = .79$, $p < .01$), support for CPEC ($r = .79$, $p < .01$), and sustainable development ($r = .71$, $p < .01$). PSD was positively and significantly correlated with support for CPEC ($r = .71$, $p < .01$) and sustainable development ($r = .67$, $p < .01$). Support for CPEC (DV) was positively and significantly correlated with sustainable development ($r = .79$, $p < .01$). These findings provide initial support for H1 and H2 and are consistent with H5 and H6.

5.3 Multiple Regression Analysis

Multiple regression analysis was conducted to test H1 and H2, with Support for CPEC as the dependent variable and PED and PSD as predictors. The model summary is reported in Table 5(a), the ANOVA results in Table 5(b), and the coefficients in Table 5(c).

The model was statistically significant, $F(2, 97) = 86.987$, $p < .001$, and accounted for 64.2% of the variance in Support for CPEC ($R^2 = .642$, adjusted $R^2 = .635$). PED was a significant positive predictor of Support for CPEC ($\beta = .612$, $t = 6.055$, $p < .001$), and PSD was also a significant positive predictor ($\beta = .222$, $t = 2.200$, $p = .030$). H1 and H2 are therefore supported. The standardized beta for PED (.612) indicates that a one-standard-deviation increase in PED is associated with a .612-standard-deviation increase in Support for CPEC, holding PSD constant. The corresponding value for PSD is .222.

Table 3. Cronbach's alpha and descriptive statistics

Variable	K	M	SD	Min	Max	α
Perceived Economic Development (PED)	11	45.52	5.57	22	55	0.87
Perceived Societal Development (PSD)	11	42.31	6.14	21	55	0.86
Support for CPEC	8	32.35	4.62	14	40	0.90
Sustainable Development (SD)	11	44.88	7.19	20	55	0.95

Note: K = number of items; M = mean; SD = standard deviation; α = Cronbach's alpha. Support for CPEC is measured with eight items, correcting an earlier inconsistency with Table 3.

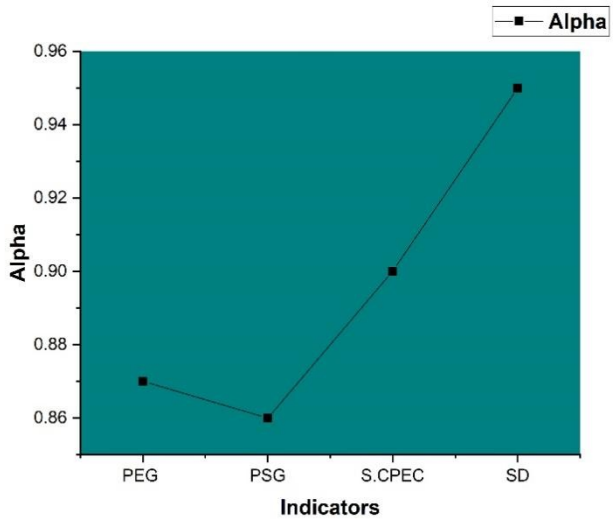


Figure 3. Alpha reliability (retain; relabel x-axis categories as PED, PSD, Support for CPEC, and SD).

Correction of a previous reporting error. An earlier draft reported "PED (F = .612, p < .001)" as a significant predictor. This was incorrect: .612 is the standardized beta coefficient for PED, not an F-statistic. The correct reporting is β = .612, t = 6.055, p < .001. The F-statistic for the overall model is 86.987 (p < .001). All statistical terminology has been checked and corrected throughout the manuscript.

Table 4. Results of correlation analysis

Variables	1	2	3	4
1. Perceived Economic Development (PED)	—	.79	.79	.71
2. Perceived Societal Development (PSD)		—	.71	.67
3. Support for CPEC			—	.79
4. Sustainable Development (SD)				—

Note: $p < 0.01$.

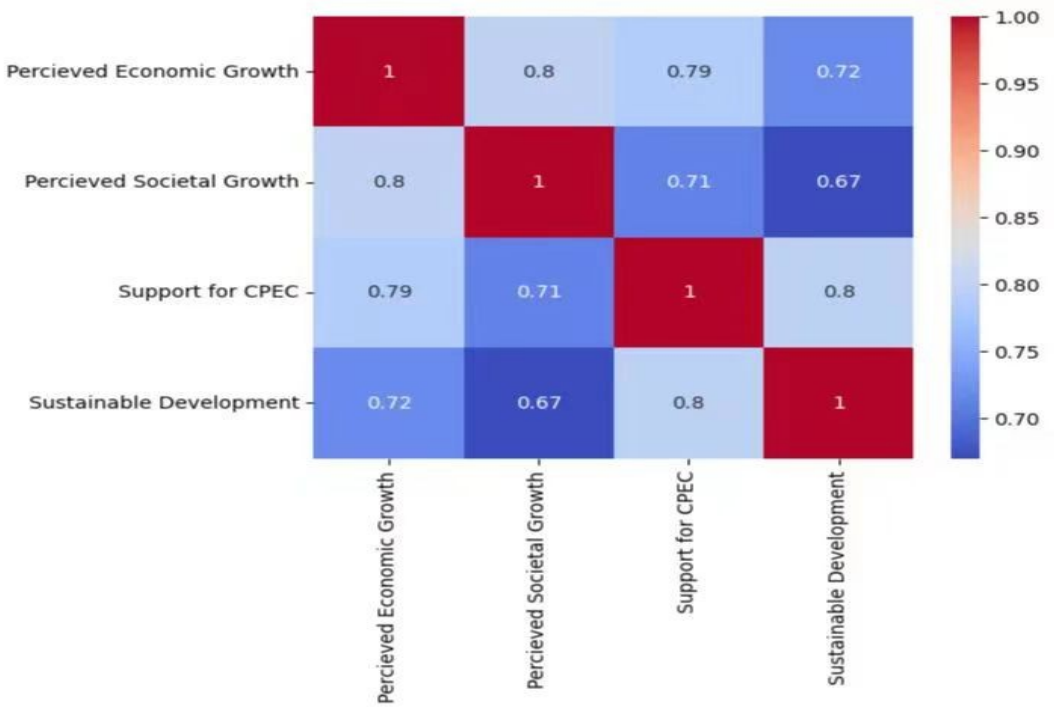


Figure 4. Correlation plot (retain; relabel variables as PED, PSD, Support for CPEC, and SD).

Table 5(a). Model summary

Model	**R**	**R^2**	**Adjusted R^2**	**Std. error of the estimate**
1	.801[a]	.642	.635	2.792

Note: [a] Predictors: (Constant), PED, PSD.

5.4 Mediation Analysis

Mediation analysis was conducted using a bootstrapping method with 5,000 bootstrapped samples, following the approach recommended by Hayes (2013), to assess the mediating effect of Sustainable Development (SD) in the relationships between PED and Support for CPEC, and between PSD and Support for CPEC. The results are reported in Table 6.

Table 5(b). ANOVA[a]

Model	**Sum of squares**	**Df**	**Mean square**	**F**	**Sig.**
1 Regression	1356.452	2	678.226	86.987	< .001
Residual	756.298	97	7.797		
Total	2112.750	99			

Note: [a] Dependent variable: Support for CPEC.

Table 5(c). Coefficients[a]

ModelUnstandardized BStd. errorStandardized betatSig.					
1 (Constant)	2.586	2.314		1.118	.266
PED	**.507**	**.084**	**.612**	**6.055**	**< .001**
PSD	**.167**	**.076**	**.222**	**2.200**	**.030**

Note: [a] Dependent variable: Support for CPEC. PED = Perceived Economic Development; PSD = Perceived Societal Development.

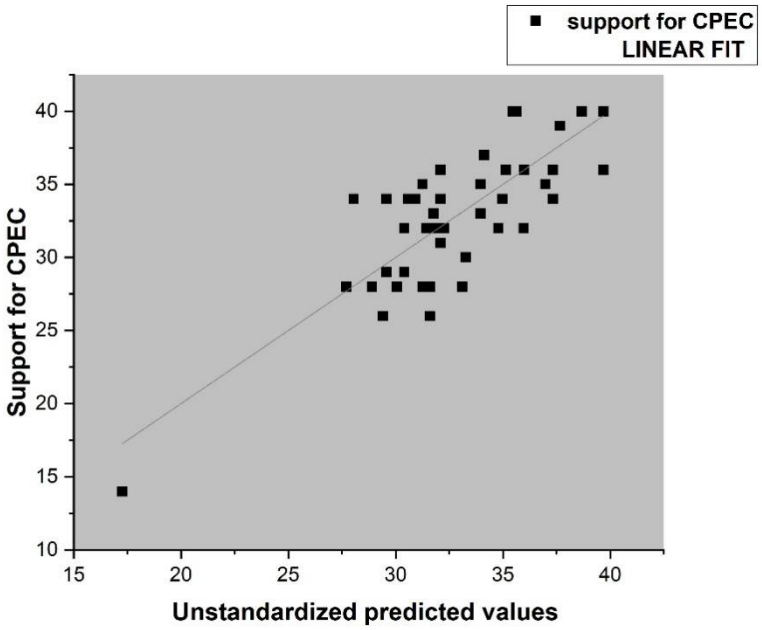


Figure 5. Linear fit model (retain)

Table 6. Mediation analysis

IV	M	DV	Effect of IV on M (a)	Effect of M on DV (b)	Direct effect (c′)	Indirect effect (a × b)	95% CI	Result
PED	SD	Support for CPEC	.71	.62	.28	.18	[0.128, 0.354]	H3 supported
PSD	SD	Support for CPEC	67	62	24	.15	[0.246, 0.494]	H4 supported

Note: IV = independent variable; M = mediator; DV = dependent variable; CI = confidence interval. PED = Perceived Economic Development; PSD = Perceived Societal Development; SD = Sustainable Development. $p < .01$.

The indirect effect of PED on Support for CPEC through SD was significant (indirect effect = .18, 95% CI [0.128, 0.354]), supporting H3. The indirect effect of PSD on Support for CPEC through SD was also significant (indirect effect = .15, 95% CI [0.246, 0.494]), supporting H4. Because the direct effects of PED and PSD on Support for CPEC remained significant after including SD in the model, the mediation is partial rather than full. These results indicate that SD partially mediates the relationships between PED and Support for CPEC and between PSD and Support for CPEC.

In summary, the empirical findings directly answer the research questions of this study. Regarding RQ1, the significant positive correlations and regression coefficients indicate that CPEC-related perceptions are strongly and positively associated with both Perceived Economic Development (PED; $\beta = 0.612$) and Perceived Societal Development (PSD; $\beta = 0.222$) among local communities. Regarding RQ2, the mediation analysis indicates that these perceived developments are significantly associated with sustainable development, which in turn is associated with overall support for the CPEC initiative.

5.5 Qualitative Findings

The qualitative findings are presented below as four themes. Anonymized participant identifiers are used (e.g., Urban-1, Rural-2, Peri-urban-3).

Theme 1: Employment quality and sustainability. Urban interviewees reported positive attitudes toward job creation and transportation infrastructure improvement. Rural respondents, however, noted that many jobs created by CPEC were short-lived and low-wage, suggesting that the corridor may support short-term economic activity but not long-term employment unless accompanied by sustainable job training and entrepreneurship programs.

"There are some construction jobs that have been going on long enough, yet where are we going to get the next job after the roads are built?" (Rural-2)

This concern relates to SDG 8, which calls for durable and quality employment opportunities, and to SDG 10, which calls for reduced inequalities (Zehri et al., 2024).

Theme 2: Service accessibility and rural–urban inequality. Several respondents reported that transport infrastructure had improved access to health and education facilities, but rural populations continued to face substantial difficulties in accessing proper services.

"There is nothing wrong with good roads, yet there are no doctors in the village, and what about it?" (Rural-4)

This finding suggests that perceived improvements in physical infrastructure do not automatically translate into improved healthcare and education services. To support SDG 3 and SDG 4, infrastructure investment may need to be combined with social investment that ensures quality and affordable services. These results are consistent with AbuShihab et al. (2024).

Theme 3: Environmental degradation. Industrial-area interviewees were aware of pollution, deforestation, and other negative effects associated with large infrastructure projects under CPEC.

These issues are directly associated with SDG 13 (Climate Action), which calls for urgent and effective steps to slow climate change. Environmental mitigation measures and sustainable practices should be considered during the implementation of CPEC to align with SDG 13. The results of the quantitative and

qualitative analyses are primarily consistent with previous studies but indicate considerable deficiencies in the effective implementation of CPEC, mainly related to sustainable development.

Theme 4: Benefit distribution and community inclusion. Several respondents noted that the economic benefits of CPEC were not equally shared. Urban respondents reported more positive perceptions of economic opportunities than rural respondents. This finding is consistent with the quantitative results reported in Section 5.3 and with prior research on geographic variation in PED (Zou et al., 2022).

6. Discussion

The quantitative findings indicate that PED and PSD are significant predictors of Support for CPEC, and that SD partially mediates these relationships. These findings are consistent with SET, which predicts that community support is a function of perceived benefits relative to perceived costs (Lehtinen & Aaltonen, 2024). The stronger beta for PED (β = .612) than for PSD (β = .222) suggests that economic considerations weigh more heavily than social considerations in shaping support for CPEC, at least in this sample. This is consistent with prior research on CPEC that has emphasized economic benefits (Kanwal et al., 2022; Shahzad et al., 2023).

The mediation results indicate that sustainable development is one mechanism through which PED and PSD are associated with support for CPEC. This finding contributes to the literature by specifying a mechanism that has rarely been tested empirically in the CPEC context.

The qualitative findings help explain these statistical relationships. Theme 1 indicates that employment quality not merely job creation matters for how communities evaluate CPEC. Theme 2 indicates that service accessibility is unevenly distributed between urban and rural areas, which may explain why PSD has a weaker association with support than PED. Theme 3 indicates that environmental concerns are salient for communities near industrial zones, which may dampen support among those respondents. Theme 4 indicates that benefit distribution is perceived as unequal, which is consistent with the moderate size of the PSD effect.

Cautions regarding causal language. Because the study is based on cross-sectional, perceptual data, the findings should be interpreted as associations rather than causal effects. Terms such as "affect," "influence," and "enhance" have been used cautiously throughout this manuscript. Similarly, perceived improvements related to education, health, employment, or the environment should not be interpreted as evidence that specific Sustainable Development Goals have been achieved. They are perceptions of improvement, not objective measures of SDG attainment.

7. Conclusion and Policy Implications

(The conclusion is retained with three changes: (a) causal language is moderated; (b) the limitations are updated to reflect the cross-sectional, perceptual design more explicitly; and (c) a final statement on the study's contribution is added.)

This study examined the perceptions of local communities toward CPEC and their association with Perceived Economic Development (PED) and Perceived Societal Development (PSD). The findings indicate that CPEC is perceived largely positively in terms of economic growth, job creation, and improved infrastructure and regional connectivity, which are relevant to SDG 8 and SDG 9. However, the analysis of benefit distribution revealed inequality, especially between urban and rural populations, which relates to SDG 10 (Reduced Inequalities). Perceived societal development was also moderate, with increased access

to education and healthcare but without concurrent investment in quality social services; this suggests that SDG 3 and SDG 4 remain only partially achieved. In addition, concerns about environmental pollution and deforestation highlight the need to incorporate SDG 13 (Climate Action) into project planning. These findings point to a need for policy coherence that emphasizes long-term, sustainable employment and training, especially in prospective CPEC regions, and for greater investment in social services, including health and education infrastructure. Environmental protections, including eco-friendly techniques and renewable energy systems, may help reduce the environmental impact of growth. Civil society engagement should also be enhanced to respond to public concerns and strengthen public trust in the CPEC program. Monitoring and evaluation of CPEC should be result-oriented and efficient to assess the economic, social, and environmental impact of the project.

Limitations. The study has several limitations. First, it was conducted in areas most affected by CPEC; it therefore does not capture the perceptions of people in areas that are less affected. Second, the cross-sectional research design limits the analysis of change in perceptions over time; longitudinal studies are needed. Third, while the quantitative interviews were informative, they were brief, and the sample size could be increased in future research. Fourth, the evaluation of impacts relies on perception only; objective assessment would provide a stronger basis for evaluation. Finally, there is a lack of generally agreed-upon standards for PSD measurement, which raises issues for comparison; this paper's findings emphasize the need for the development of standard metrics in the field.

In direct response to the research questions, this study concludes that CPEC is perceived to be positively associated with both Perceived Economic Development (PED) and Perceived Societal Development (PSD) among local communities, as evidenced by the regression coefficients ($\beta = 0.612$ for PED; $\beta = 0.222$ for PSD) and significant correlations observed in the empirical analysis. Furthermore, the mediation results indicate that these perceived developments are significantly associated with sustainable development, which in turn is associated with community support for the CPEC initiative. However, the qualitative insights reveal that while perceptions are largely favorable, the realization of long-term sustainable outcomes may require addressing persistent challenges related to equitable benefit distribution, rural service accessibility, and environmental protection. This study thus contributes to the existing literature by providing empirical evidence that community perceptions serve as critical indicators for evaluating the sustainability of mega-infrastructure projects, and it offers practical guidance for policymakers seeking to align large-scale development initiatives with the Sustainable Development Goals.

Conflict of interest Statement

The authors declare that they have no conflict of interest.

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References

Note: The Sohail & Chen (2022) entry has been removed because the source was retracted. A note on this removal is included below.

Abbas, J., Mamirkulova, G., Al-Sulaiti, I., Al-Sulaiti, K. I. & Dar, I. B. 2025. Mega-infrastructure development, tourism sustainability and quality of life assessment at world heritage sites: catering to COVID-19 challenges. Kybernetes, 54, 1993–2018.

Abushihab, K., Obaideen, K., Alameddine, M., Alkurd, R. A. F., Khraiwehs, H. M., Mohammad, Y., Abdelrahim, D. N., Madkour, M. I. & Faris, M. E. 2024. Reflection on Ramadan fasting research related to sustainable development goal 3 (Good Health and Well-Being): a bibliometric analysis. Journal of Religion and Health, 63, 3329–3359.

Ahmad, L., Haider, A., Ahmad, I. & Jawad, A. 2025. Can CPEC Unlock Pakistan's Economic Potential and Overcome Its Challenges? Journal of Asian Development Studies, 14, 1361–1372.

Alam, S. & Iqbal, K. 2025. Contours of connectivity: Socioeconomic Transformation and Local Perception of China-Pakistan Economic Corridor (CPEC) in upper Hunza, Pakistan. FWU Journal of Social Sciences, 19.

Ali, L., Mi, J., Shah, M., Shah, S. J., Khan, S., Ullah, R. & Bibi, K. 2018. Local residents' attitude towards road and transport infrastructure (a case of China Pakistan economic corridor). Journal of Chinese Economic and Foreign Trade Studies, 11, 104–120.

Anwar, S. U., Wuyi, Z., Ali Shah, S. Z., Ullah, Q., Amir, S. M. & Syed, A. 2022. The resilient economic impact of CPEC and future of MNCs: Evidence from Pakistan. Frontiers in Environmental Science, 10, 912975.

Anwar, S. U., Yuan, P. Z., Wuyi, Z., Amir, S. M., Rehman, S. U., Yang, L. & Shah, S. Z. A. 2024. Nexus among the perceived infrastructural, social, economic, and environmental impact of CPEC: A case of Pakistan. Heliyon, 10.

Arshad, H. M., Akhtar, M. W., Imran, M., Batool, I., Asrar-Ul-Haq, M. & Akbar, M. 2024. Effectiveness of leader-member exchange on employee outcomes in cross-cultural environment of China–Pakistan economic corridor. Kybernetes, 53, 1209–1228.

Asif, M., Elmarakby, E. H., Sarhan, N., Hashim, M., Zia, M. A., Bhatti, U. A. & Bhatti, M. A. 2025. Exploring the interconnection of emotional intelligence and transformational leadership in enhancing CPEC project success. Acta Psychologica, 261, 105897.

Asif, M., Khan, M. I. & Pandey, A. 2024. Navigating the inclusive and sustainable energy transitions in South Asia: Progress, priorities and stakeholder perspectives. Energy Conversion and Management, 313, 118589.

Aslam, M., Elahi, M. E., Ahmed, N., Waqar, M. & Wali, S. 2023. Impact of CPEC on Social Welfare and its Significance on different Provinces of Pakistan. Indus Journal of Social Sciences, 1, 1–17.

Azhar, A., Abutabenjeh, S. & Nukpezah, J. 2025. Budgeting system and economic development outcomes. Public Administration Quarterly, 49, 106–124.

Aziz, N., Baber, J., Raza, A. & He, J. 2025. Feminist-environment nexus: a case study on women's perceptions toward the China-Pakistan economic corridor and their role in improving the environment. Journal of Environmental Planning and Management, 68, 363–385.

Balsalobre-Lorente, D., Abbas, J., He, C., Pilař, L. & Shah, S. A. R. 2023. Tourism, urbanization and natural resources rents matter for environmental sustainability: The leading role of AI and ICT on sustainable development goals in the digital era. Resources Policy, 82, 103445.

Bashir, S., Noreen, S. & Awan, M. B. 2022. China-Pakistan Economic Corridor And Collaboration In Higher Education: Trends, Perspectives And Prospects. Journal of Positive School Psychology, 6.

Becker, S. O. & Woessmann, L. 2024. Education and Socioeconomic Development During the Industrialization. Springer Books, 459–479.

Bensadi, A. 2024. Assessing the impact of renewable energy integration on energy efficiency within the China-Pakistan economic corridor (CPEC). Scientific Reports, 14, 29374.

Braun, V. & Clarke, V. 2006. Using thematic analysis in psychology. Qualitative Research in Psychology, 3, 77–101.

Cohen, J. 1992. A power primer. Psychological Bulletin, 112, 155–159.

Crisp, R., Waite, D., Green, A., Hughes, C., Lupton, R., Mackinnon, D. & Pike, A. 2024. 'Beyond GDP' in cities: Assessing alternative approaches to urban economic development. Urban Studies, 61, 1209–1229.

Editors, P. O. 2024. Retraction: Public perceptions About Foreign Investment, A PLS-SEM Analysis toward republic sustainable infrastructure. Public Library of Science San Francisco, CA USA.

George, D. & Mallery, P. 2024. IBM SPSS statistics 29 step by step: A simple guide and reference, Routledge.

Green, S. B. 1991. How many subjects does it take to do a regression analysis? Multivariate Behavioral Research, 26, 499–510.

Hayes, A. F. 2013. Introduction to mediation, moderation, and conditional process analysis: A regression-based approach. Guilford Press.

Homans, G. C. 1958. Social behavior as exchange. American Journal of Sociology, 63, 597–606.

Hussain, A., Mandić, A. & Fusté-Forné, F. 2024a. Transforming communities: Analyzing the effects of infrastructure and tourism development on social capital, livelihoods, and resilience in Gilgit-Baltistan, Pakistan. Journal of Hospitality and Tourism Management, 59, 276–295.

Hussain, M., Jamali, A. B., Nisar, R. D. & Omar, A. 2024b. The China-Iran strategic deal and CPEC: Navigating the influence of pragmatic balancing in China's relations with Iran and Pakistan. Politics & Policy, 52, 227–244.

Hussain, Z., Ali, B., Ul Hassan, M. & Ali, M. 2025. Environmental Concerns of China-Pakistan Economic Corridor (CPEC) on Gilgit Baltistan. THE ASIAN BULLETIN OF GREEN MANAGEMENT AND CIRCULAR ECONOMY, 5, 110–125.

Islam, H. 2025. Nexus of economic, social, and environmental factors on sustainable development goals: The moderating role of technological advancement and green innovation. Innovation and Green Development, 4, 100183.

Kanwal, S., Chong, R. & Pitafi, A. H. 2019. China-Pakistan economic corridor projects development in Pakistan: Local citizens benefits perspective. Journal of Public Affairs, 19, e1888.

Kanwal, S., Pitafi, A. H., Rasheed, M. I., Pitafi, A. & Iqbal, J. 2022. Assessment of residents' perceptions and support toward development projects: A study of the China-Pakistan Economic Corridor. The Social Science Journal, 59, 102–118.

Kanwal, S., Rasheed, M. I., Pitafi, A. H., Pitafi, A. & Ren, M. 2020. Road and transport infrastructure development and community support for tourism: The role of perceived benefits, and community satisfaction. Tourism Management, 77, 104014.

Karim, A. 2022. [Insert full bibliographic details for the Karim (2022) source cited in Section 4.2.]

Khan, A., Ilmas, F., Zubair, M., Khan, A. & Zhong, L. H. 2022. The impact of CPEC on the economy of Pakistan. Journal of Positive School Psychology, 6, 4214–4223.

Lee, C.-K., Kim, J. S. & Kim, J. S. 2018. Impact of a gaming company's CSR on residents' perceived benefits, quality of life, and support. Tourism Management, 64, 281–290.

Lehtinen, J. & Aaltonen, K. 2024. Community stakeholders' online engagement in infrastructure projects: a theory-testing single-case study. Construction Management and Economics, 42, 758–786.

McLaughlin, M. 2024. The geoeconomics of belt and road disputes: A case study on the China-Pakistan Economic Corridor. Asian Journal of International Law, 14, 94–122.

Shahzad, A. & Sunawar, L. 2024. Challenges and Outcomes of China-Pakistan Economic Corridor (CPEC): A Narrative Review. Journal of Political Stability Archive, 2, 117–128.

Shahzad, N., Khuram, H., Khan, H., Bajwa, M. & Nawaz, S. 2023. Opportunities And Challenges Of The China-Pakistan Economic Corridor (Cpec): A Game Changer In South Asian Countries And Impact Of Cpec On Pakistani Economy. Journal of Positive School Psychology, 7, 860–879.

\~\~Sohail, M. T. & Chen, S. 2022. Public perceptions About Foreign Investment, A PLS-SEM Analysis toward republic sustainable infrastructure. Plos one, 17, e0277673.\~\~ [REMOVED — RETRACTED]

Sulaiman, S. 2024. The CPEC and SDGs in Pakistan: Measuring the Impact on Common Lives, Springer Nature.

Ullah, H. A., Kainaat, F., Wajid, H. & Sarwar, M. 2024. Analyzing the impact of the China-Pakistan Economic Corridor (CPEC) on Pakistani economy and society. Al-Mahdi Research Journal (MRJ), 5, 769–780.

Waheed, A., Kousar, S., Khan, M. I. & Fischer, T. B. 2024. Environmental governance in Pakistan: Perspectives and implications for the China-Pakistan economic corridor plan. Environmental and Sustainability Indicators, 23, 100443.

Yuanchun, L., Yang, S., Fei, W. & Guangsu, Z. 2024. Construction and Promotion of the Demonstration Zone for Common Prosperity. Understanding China's New Common Prosperity: Significance, Connotations, and Goals. Springer.

Zehri, C., El Amin, B. M., Inaam, Z. & Sekrafi, H. 2024. Exploring the nexus of decent work, financial inclusion, and economic growth: A study aligned with SDG 8. Sustainable Futures, 7, 100213.

Zhang, C. 2023. China's Infrastructure Construction in the Middle East. Routledge Companion to China and the Middle East and North Africa. Routledge.

Zhang, F., Xiao, F. & Yu, P. 2024. A model of sustainable development in economic, social, and environmental aspects: The role of social capital in China. Environmental Research Communications, 6, 045001.

Zhang, L., Huang, F., Lu, L. & Ni, X. 2021. Green financial development improving energy efficiency and economic growth: a study of CPEC area in COVID-19 era. IRTG 1792 Discussion Paper.

Zou, L., Shen, J. H., Zhang, J. & Lee, C. C. 2022. What is the rationale behind China's infrastructure investment under the Belt and Road Initiative. Journal of Economic Surveys, 36, 605–633.